JACOB JIBU

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SUMMARY

Electrical and Electronics Engineering undergraduate with hands-on experience in embedded systems, industrial automation, hardware validation, firmware development, wireless communication, and control systems. Experienced in leading multidisciplinary engineering projects involving ESP32-based embedded platforms, sensor interfacing, IoT systems, and machine-learning-based control applications. Skilled in system design, troubleshooting, prototype development, testing, and subsystem integration. Seeking opportunities in embedded systems, industrial automation, electrical engineering, and control systems.

EDUCATION

Vellore Institute of Technology ***Chennai, TN | 08/2023 - Present***

B.Tech in Electrical and Electronics Engineering (7th Semester) — Cumulative GPA: 8.19 / 10

Profile Focus: Excelled in Core Electrical Systems Coursework

MTHSS, Kaloor ***Ernakulam, Kerala| 07/2021- 03/2023***

Higher Secondary Education (Overall Marks: 94.66%)

SKILLS

- **Programming:** Embedded C, C, Python, Java
- **Embedded Platforms:** ESP32, ESP8266, Arduino
- **Software & Development Tools:** MATLAB, Simulink, LTspice, Keil µVision, Arduino IDE
- **Core Engineering Skills:** Embedded Systems, Circuit Design, Hardware Validation, Sensor Interfacing, IoT, Power Electronics, Control Systems.
- **Soft Skill:** Teamwork, Problem Solving, Leadership, Communication, Adaptability.
- **Languages:** English(Full professional proficiency), Malayalam(Native), Tamil(Conversational proficiency), Hindi(Listening Proficiency).

EXPERIENCE

Team IGNITION ***Chennai, TN***

Recovery Systems Engineer ***Aug 2025 – May 2026***

- Led the design and development of a controllable paraglider-based rocket recovery system.
- Coordinated hardware development, subsystem integration, testing, and reliability analysis activities.
- Conducted feasibility studies and design reviews to improve recovery system performance.
- Collaborated with multidisciplinary teams to meet design objectives and testing deadlines.
- Managed timelines, allocated responsibilities, and documented design procedures.

Apollo Tyres Ltd. ***Kalamassery, Kerala***

Automation & Industrial Intern ***Jun 2025 – Jul 2025***

- Gained hands-on exposure to industrial automation, PLC-controlled manufacturing systems, and electrical distribution networks.
- Assisted in understanding induction motors, electrical drives, industrial control panels, and preventive maintenance procedures.
- Observed electrical equipment troubleshooting, industrial safety standards, and manufacturing operations.
- Analyzed production workflows, process automation, and manufacturing efficiency in a high-volume industrial environment.

PROJECTS

Adaptive Matrix LED Headlamp System ***Jan 2026 - Apr 2026***

- Developed an Arduino-based Adaptive Matrix LED Headlamp using a multiplexed LED matrix to reduce digital I/O usage.
- Integrated YOLO-based computer vision for real-time vehicle detection and zone-based adaptive dimming.
- Developed embedded control logic for real-time LED control, hardware integration, and automotive lighting.

Technologies: Arduino • Embedded C • YOLO • Computer Vision • LED Matrix • PWM

ESP32-Based ML Smart Energy Management & Predictive Load Shedding System ***Jul 2025 - Dec 2025***

- Developed an ESP32-based IoT energy management system using Machine Learning for electrical load forecasting.
- Implemented MQTT communication for real-time monitoring and mobile-controlled load shedding.
- Developed embedded firmware for relay control, data acquisition, and automated power management.

Technologies: ESP32 • Embedded C • Machine Learning • MQTT • IoT • Relay Control

Smart Modular Climate Regulation Platform (SIH Internal Prototype) ***Aug 2025 – Oct 2025***

- Developed an ESP32-based IoT climate control system for agricultural cold storage.
- Integrated temperature & humidity sensors, Peltier cooling, and automated ventilation for environmental regulation.
- Developed embedded firmware with MQTT communication for sensor interfacing, actuator control, and mobile monitoring.

Technologies: ESP32 • Embedded C • MQTT • IoT • Peltier Module • Temperature & Humidity Sensors

Hack Computer Architecture System ***Aug 2026 – Present***

- Building a 16-bit computer from basic logic gates based on the Hack architecture.
- Implementing the CPU, memory, clock generation, and display subsystems.
- Exploring computer architecture and digital system design concepts.

Extracurricular Activities

Team IGNITION, VIT Chennai

- Participated in the design and development of experimental rocketry systems.
- Worked on recovery mechanisms, wireless communication systems, and embedded system integration.

Smart India Hackathon (Internal Selection Round)

- Participated in the development of an IoT-based climate regulation system for agricultural applications.

Embedded Systems and Electronics Projects

- Designed and developed multiple projects involving ESP32, IoT, machine learning, and computer architecture.